

# AWS EC2 Linux Customization

## Instance Setup

- Launched an AWS EC2 instance with Amazon RedhatLinux 9.
- Connected to the instance using SSH.

## Software Installation

- Installed MySQL 8.0.36.

## Security Configuration

- Configured firewall rules (Security Groups) to allow SSH access (port 22) and MySQL access (port 3306) only from trusted sources.

## File Management

- Moved project files to `/home/ec2-user`

## MySQL Setup

- Installed and configured MySQL 8.0.36.
- Created databases and users, imported data from CSV files, and executed SQL queries as detailed earlier.

## System Updates

- Ran `sudo yum update` to ensure all packages are up to date.

## AWS EC2 Instance Details

- **Instance ID:** i-048f9bdc89aae561c
- **Public IPv4 Address:** 35.160.43.118
- **Private IPv4 Address:** 172.31.31.155
- **Platform:** Red Hat Enterprise Linux 9
- **Instance Type:** t2.micro
- **AMI ID:** ami-0423fca164888b941
- **Launch Time:** Wed Jul 03 2024 09:03:20 GMT
- **AWS Region:** us-west-2

# MySQL Operations Summary

## 1. Add MySQL Service to Startup:

```
sudo systemctl enable mysql
```

## 2. Make MySQL Command Accessible:

Ensure MySQL binaries are in system's PATH. If not, add it manually or edit .bashrc or .bash\_profile.

```
export PATH=$PATH:/path/to/mysql/bin
```

## 3. Reset Password of Root User:

- Start MySQL in safe mode with skip-grant-tables option to bypass authentication.

```
sudo mysqld_safe --skip-grant-tables &
```

Update the root password.

```
USE mysql;
```

```
UPDATE user SET Password=PASSWORD('new_password') WHERE User='root';  
FLUSH PRIVILEGES;
```

- Exit MySQL and restart normally.

```
sudo systemctl restart mysql
```

## 4. Create Database 'hms\_training':

```
CREATE DATABASE hms_training;
```

## 5. Create User and Grant Permissions: Replace 'username' and 'password' with actual values.

```
CREATE USER 'your_username'@'localhost' IDENTIFIED BY 'my_password';  
GRANT ALL PRIVILEGES ON hms_training.* TO 'my_username'@'localhost';  
FLUSH PRIVILEGES;
```

## 6. Create TRANSACTION\_DETAIL Table and Load Data: Use provided SQL script and CSV dump to create and load data into TRANSACTION\_DETAIL table.

## 7. Write Query for Specific Result Set:

```
SELECT SUM(CREDIT_AMOUNT) AS Total_Credit_Amount
FROM TRANSACTION_DETAIL
WHERE SOURCE_ADDRESS = 'cargills'
  AND SUBSCRIBER_TYPE = 'POST'
  AND OBJECT_TYPE = 'hsenidmobile.mrecharge.model.IPGRemoteMerchantTransactionDetail'
  AND STATUS = 3
  AND REMOTE_TRX_CREATED_TIME BETWEEN '2011-05-01' AND '2012-12-31';
```

#### 8. Optimize Query Execution Time:

- Analyze the query execution plan (EXPLAIN) to identify inefficient parts.
- Create indexes on columns used in WHERE clause (SOURCE\_ADDRESS, SUBSCRIBER\_TYPE, OBJECT\_TYPE, STATUS, REMOTE\_TRX\_CREATED\_TIME).

#### 9. Create Temporary Table and Insert Data:

```
CREATE TEMPORARY TABLE temp_transaction_detail
SELECT *
FROM TRANSACTION_DETAIL
LIMIT 200;
```

#### 10. Identify Configuration Parameters and Show Variables:

- Edit my.cnf to set innodb\_buffer\_pool\_size = 512M.
- Adjust innodb\_log\_file\_size accordingly.
- Restart MySQL for changes to take effect.
- Use SHOW VARIABLES; in MySQL to display all configuration parameters

## Here are some outputs of the script

```
ec2-user@ip-172-31-31-155:~$  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> USE hms_training;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables in hms_training |  
+-----+  
| TRANSACTION_DETAIL |  
+-----+  
1 row in set (0.00 sec)  
  
mysql> DESCRIBE TRANSACTION_DETAIL;  
+-----+  
--+  
| Field | Type | Null | Key | Default | Extra |  
+-----+  
--+  
| transaction_id | int | NO | PRI | NULL | auto_increment |  
| transaction_date | date | YES | | NULL | |  
| amount | decimal(10,2) | YES | | NULL | |  
| description | varchar(255) | YES | | NULL | |  
| SOURCE_ADDRESS | varchar(255) | YES | MUL | NULL | |  
| SUBSCRIBER_TYPE | varchar(255) | YES | | NULL | |  
| OBJECT_TYPE | varchar(255) | YES | | NULL | |  
| STATUS | int | YES | | NULL | |  
| REMOTE_TRX_CREATED_TIME | datetime | YES | | NULL | |  
+-----+  
--+  
9 rows in set (0.00 sec)  
  
ec2-user@ip-172-31-31-155:~$  
Authenticating with public key "imported-openssh-key"  
Register this system with Red Hat Insights: insights-client --register  
Create an account or view all your systems at https://red.ht/insights-dashboard  
Last login: Wed Jul 3 12:42:07 2024 from 123.231.84.136  
[ec2-user@ip-172-31-31-155 ~]$ sudo systemctl enable mysqld  
sudo systemctl is-enabled mysqld  
enabled  
[ec2-user@ip-172-31-31-155 ~]$ echo 'export PATH=$PATH:/usr/local/mysql/bin' >>  
~/.bash_profile  
source ~/.bash_profile  
which mysql  
mysql --version  
/usr/bin/mysql  
mysql Ver 8.0.36 for Linux on x86_64 (Source distribution)  
[ec2-user@ip-172-31-31-155 ~]$ SHOW DATABASES;  
-bash: SHOW: command not found  
[ec2-user@ip-172-31-31-155 ~]$ CREATE DATABASE hms_training;  
SHOW DATABASES;  
-bash: CREATE: command not found  
-bash: SHOW: command not found  
[ec2-user@ip-172-31-31-155 ~]$ mysql -u Fildon23 -p  
Enter password:  
ERROR 1045 (28000): Access denied for user 'Fildon23'@'localhost' (using password: YES)  
[ec2-user@ip-172-31-31-155 ~]$ mysql -u Fildon23 -p  
Enter password:  
Welcome to the MySQL monitor. Commands end with ; or \g.  
Your MySQL connection id is 9  
Server version: 8.0.36 Source distribution  
  
Copyright (c) 2000, 2024, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| hms_training |  
| information_schema |  
| performance_schema |  
+-----+  
3 rows in set (0.01 sec)  
  
mysql>
```

```
ec2-user@ip-172-31-31-155:~  
  
| version_compile_os | | Linux |  
  
| version_compile_zlib | | 1.2.13 |  
  
| wait_timeout | | 300 |  
  
| warning_count | | 0 |  
  
| windowing_use_high_precision | | ON |  
  
| xa_detach_on_prepare | | ON |  
  
+-----+  
664 rows in set (0.00 sec)  
  
mysql> SHOW VARIABLES LIKE 'innodb_buffer_pool_size';  
+-----+  
| Variable_name | Value |  
+-----+  
| innodb_buffer_pool_size | 134217728 |  
+-----+  
1 row in set (0.00 sec)  
  
mysql> SHOW VARIABLES LIKE 'innodb_log_file_size';  
+-----+  
| Variable_name | Value |  
+-----+  
| innodb_log_file_size | 268435456 |  
+-----+  
1 row in set (0.01 sec)  
  
mysql>
```

```
ec2-user@ip-172-31-31-155:~  
664 rows in set (0.00 sec)  
  
mysql> SHOW VARIABLES LIKE 'innodb_buffer_pool_size';  
+-----+  
| Variable_name | Value |  
+-----+  
| innodb_buffer_pool_size | 134217728 |  
+-----+  
1 row in set (0.00 sec)  
  
mysql> SHOW VARIABLES LIKE 'innodb_log_file_size';  
+-----+  
| Variable_name | Value |  
+-----+  
| innodb_log_file_size | 268435456 |  
+-----+  
1 row in set (0.01 sec)  
  
mysql> SHOW INDEX FROM TRANSACTION_DETAIL;  
ERROR 1046 (3D000): No database selected  
mysql> USE hms_training  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
mysql> SHOW INDEX FROM TRANSACTION_DETAIL;  
+-----+  
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible |  
+-----+  
| TRANSACTION_DETAIL | 0 | PRIMARY | 1 | transaction_id | A | 4 | NULL | NULL | | BTREE | | | YES |  
| TRANSACTION_DETAIL | 1 | idx_composite | 1 | SOURCE_ADDRESS | A | 1 | NULL | NULL | YES | BTREE | | | YES |  
| TRANSACTION_DETAIL | 1 | idx_composite | 2 | SUBSCRIBER_TYPE | A | 1 | NULL | NULL | YES | BTREE | | | YES |  
| TRANSACTION_DETAIL | 1 | idx_composite | 3 | OBJECT_TYPE | A | 1 | NULL | NULL | YES | BTREE | | | YES |  
| TRANSACTION_DETAIL | 1 | idx_composite | 4 | STATUS | A | 1 | NULL | NULL | YES | BTREE | | | YES |  
| TRANSACTION_DETAIL | 1 | idx_composite | 5 | REMOTE_TRX_CREATED_TIME | A | 1 | NULL | NULL | YES | BTREE | | | YES |  
+-----+  
6 rows in set (0.01 sec)  
  
mysql>
```

```
ec2-user@ip-172-31-31-155:~$ mysql
mysql> SHOW INNODB LOGS;
+-----+
| Variable_name | Value |
+-----+
| innodb_log_file_size | 268435456 |
+-----+
1 row in set (0.01 sec)

mysql> SHOW INDEX FROM TRANSACTION DETAIL;
ERROR 1046 (3D000): No database selected
mysql> USE hms_training;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW INDEX FROM TRANSACTION DETAIL;
+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible |
+-----+
| TRANSACTION_DETAIL | 0 | PRIMARY | 1 | transaction_id | A | 4 | NULL | NULL | NULL | BTREE | | | YES |
| TRANSACTION_DETAIL | 1 | idx_composite | 1 | SOURCE_ADDRESS | A | 1 | NULL | NULL | YES | BTREE | | | YES |
| TRANSACTION_DETAIL | 1 | idx_composite | 2 | SUBSCRIBER_TYPE | A | 1 | NULL | NULL | YES | BTREE | | | YES |
| TRANSACTION_DETAIL | 1 | idx_composite | 3 | OBJECT_TYPE | A | 1 | NULL | NULL | YES | BTREE | | | YES |
| TRANSACTION_DETAIL | 1 | idx_composite | 4 | STATUS | A | 1 | NULL | NULL | YES | BTREE | | | YES |
| TRANSACTION_DETAIL | 1 | idx_composite | 5 | REMOTE_TRX_CREATED_TIME | A | 1 | NULL | NULL | YES | BTREE | | | YES |
+-----+
6 rows in set (0.01 sec)

mysql> SELECT * FROM TRANSACTION_DETAIL LIMIT 10;
+-----+
| transaction_id | transaction_date | amount | description | SOURCE_ADDRESS | SUBSCRIBER_TYPE | OBJECT_TYPE | STATUS | REMOTE_TRX_CREATED_TIME |
+-----+
| 1 | 2024-07-01 | 100.50 | Purchase of goods | NULL | NULL | NULL | NULL | NULL |
| 2 | 2024-07-02 | 75.20 | Payment received | NULL | NULL | NULL | NULL | NULL |
| 3 | 2024-07-03 | 150.75 | Expense reimbursement | NULL | NULL | NULL | NULL | NULL |
| 2024 | 0000-00-00 | 0.00 | NULL | NULL | NULL | NULL | NULL | NULL |
+-----+
4 rows in set (0.00 sec)

mysql>
```